

L3: How to Measure Degree Correlations

One way to quantify the degree correlations in a network is by modeling (*i.e.*, *approximating*) the relationship between the average nearest neighbor degree $k_{nn}(k)$ and the degree k with a power-law of the form:

$$k_{nn}(k) \approx a \cdot k^\mu$$

Then, we can estimate the exponent μ from the data.

- 1) If $\mu > 0$, we say that the network is **Assortative**: higher-degree nodes tend to have higher-degree neighbors and lower-degree nodes tend to have lower-degree neighbors. Think of celebrities dating celebrities, and loners dating other loners.
- 2) If $\mu < 0$, we say that the network is **Disassortative**: higher-degree nodes tend to have lower-degree neighbors. Think of a computer network in which high-degree aggregation switches connect mostly to low-degree backbone routers.
- 3) If μ is statistically not significantly different from zero, we say that the network is **Neutral**.

NOTE: There is an errata in the online Barabasi textbook about this topic where the direction of assortative and disassortative are reversed. The interpretation of degree correlations on this lecture slide is correct.

Food for Thought

Suppose that instead of this power-law relation between $k_{nn}(k)$ and k we had used a linear statistical model. How would you quantify degree correlations in that case?

Hint: How would you apply Pearson's correlation metric to quantify the correlation between degrees of adjacent nodes?